INCH POUND

MIL-C-3098/84C
10 April 1995
SUPERSEDING
MIL-C-3098/84B(USAF)
5 February 1979

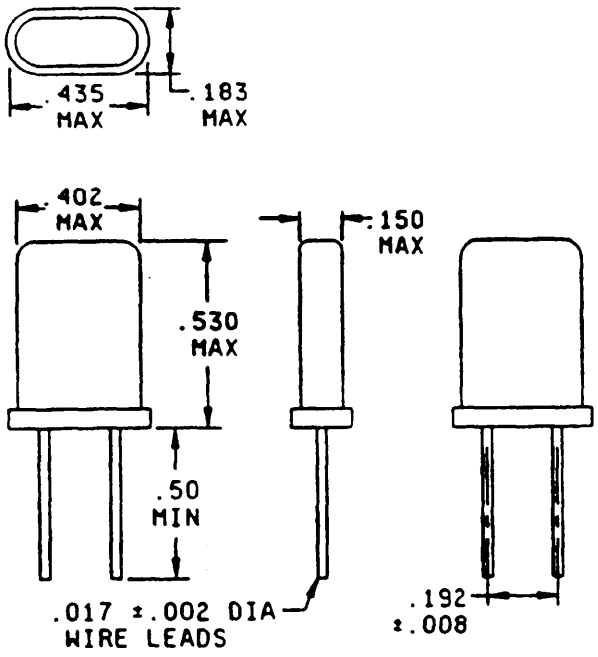
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-108/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 50 MHz to 93 MHz; fifth overtone; controlled; series resonance.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR-108/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-26/U (MIL-H-10056/13) or HC-43/U (MIL-H-10056/30-2) or HC-49/U (MIL-H-10056/36).

Dimensions, marking, and configuration: See figure 1. 1/

Frequency range: 50 MHz to 93 MHz, inclusive.

Capacitance, shunt: 5.5 pF, maximum.

Frequency stability: ±2.5 ppm.

Mode of oscillation: Fifth mechanical overtone.

Operating temperature range (controlled): +80°C to +90°C, inclusive.

Frequency tolerance: ±10 ppm (including frequency stability).

Equivalent resistance: 50 ohms maximum.

Operable: -55°C to +80°C and +90°C to +105°C

Reference temperature: +85°C ±1°C.

Drive level, rated: 2.0 ±0.4 mW.

Unwanted mode: Method II, the minimum unwanted mode effective resistance shall be at least twice that of the main mode, or 50 ohms, whichever is greater.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration high frequency: MIL-STD-202, method 204, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±3 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

1/ These crystal units where previously fabricated using Air Force Technical Order 12-1-20 and/or Collins drawing numbers 289-2844 through 2858-00.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Air Force - 85

Review activity:
Air Force - 99

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0680-23)

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